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$SO_3 \rightleftharpoons SO_2 + \frac{1}{2} O_2$

$K_2 = \frac{[SO_2][O_2]^{1/2}}{[SO_3]} \quad \therefore K_2 = \frac{1}{\sqrt{K_1}}$

option (c) is correct

21. option (b)

22. $A + B \rightleftharpoons C$

$$K_C = \frac{[C]}{[A][B]}$$

if [A] and [B] are doubled, [C] increases 4 times to maintain K_C as constant.

$\therefore$ equilibrium constant will remain the same – option (d)

23. on cooling, reverse reaction predominates and the solution is pink in colour.

$\therefore$ decrease in temperature, favours the reverse reaction ie reverse reaction is exothermic and for the forward reaction is endothermic ($\Delta H > 0$)

option (a)

24. $K_1 = \frac{[NH_3]^2}{[N_2][H_2]^3}$

$K_2 = \frac{[NO]^2}{[N_2][O_2]}$

$K_3 = \frac{[H_2O]}{[H_2][O_2]^{1/2}}$

$K = \frac{[NO]^2 [H_2O]^3}{[NH_3]^2 [O_2]^{5/2}}$

$$K = \frac{(K_2)(K_3)^3}{(K_1)} = \frac{[NO]^2}{[N_2][O_2]} \times \frac{[H_2O]^3}{[H_2]^3 [O_2]^{3/2}} \frac{[N_2][H_2]^3}{[NH_3]^2}$$

$$= \frac{[NO]^2 [H_2O]^3}{[NH_3]^2 [O_2]^{5/2}}$$

option (c)

25. Given that $K_p = 1.6$ atm

$V_1 = 20$ L $\quad V_2 = ?$

$T_1 = 400$ K $\quad T_2 = 400$ K

$K_p = P_{CO_2}$

$\therefore P_{CO_2} = 1.6$ atm

$P_1 = 0.4$ atm. $\quad P_2 = 1.6$ atm

$$\frac{P_1V_1}{T_1} = \frac{P_2V_2}{T_2}$$

$$\therefore V_2 = \frac{P_1V_1}{T_1} \times \frac{T_2}{P_2} = \frac{0.4 \text{ atm} \times 20 \text{ L}}{\cancel{400 \text{ K}}} \times \frac{\cancel{400 \text{ K}}}{1.6 \text{ atm}}$$

$= 5$L (option b)

31. $K_C = \frac{[AB]^2}{[A_2][B_2]}$

A – green

B – blue

Given that 'V' is constant (closed system)

At equilibrium

$$K_C = \frac{\left(\frac{4}{V}\right)^2}{\left(\frac{2}{V}\right)\left(\frac{2}{V}\right)} = \frac{16}{4} = 4$$

$K_p = K_C (RT)^{\Delta n}$

$K_p = 4(RT)^o = 4$

At Stage 'x'

$$Q = \frac{\left(\frac{6}{V}\right)^2}{\left(\frac{2}{V}\right)\left(\frac{1}{V}\right)} = \frac{36}{2} = 18$$

$Q > K_C$ ie., reverse reaction is favoured

At Stage 'y'

$$Q = \frac{\left(\frac{3}{V}\right)^2}{\left(\frac{3}{V}\right)\left(\frac{3}{V}\right)} = \frac{9}{3 \times 3} = 1$$

$K_C > Q$ ie., forward reaction is favoured.